

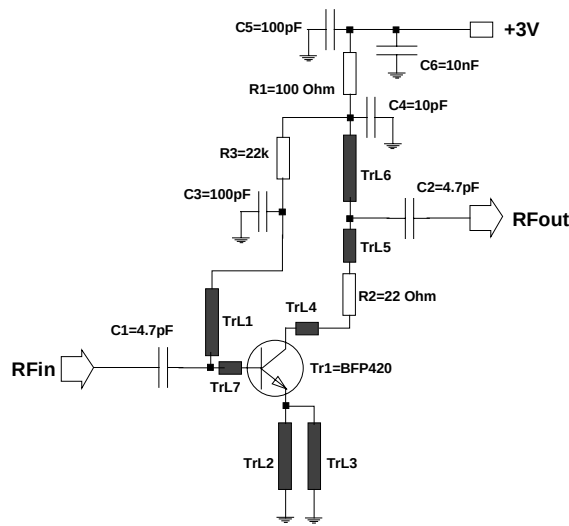
G. Lohninger

A Low-Noise-Amplifier at 900 MHz using SIEGET[®] BFP420

This application note describes a low noise amplifier at 900MHz using SIEMENS SIEGET[®] 25 BFP420. The design emphasis has been on achieving a low noise figure. A circuit description, schematic, pcb layout and components list are shown below together with measured performance data.

Data at 0.9 GHz (3V and 5mA):

Gain=18 dB / IP3out=10dBm / NF=1.35dB / R_{Lin-out} >10dB



This amplifier at 900 MHz has been realized by using microstrip lines as matching elements. The design offers a good compromise between high IP3in- values, low noise figure and high return loss.

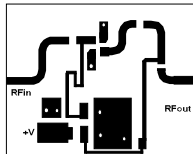
In order to optimise the design for a particular application please observe the following points:

- The layout size can be reduced by using chip inductors instead of the microstrip lines TrL1 and TrL6.
- Improved stabilization behaviour versus temperature and reduced variation in amplifier performance due to the device's Beta (current gain) distribution can be achieved by using an active bias circuit. Such a circuit is available as a single device from Siemens - BCR400. For further information please refer to Application Note 014. However the resistors R1 and R3 are sufficient in most applications for stabilization purposes.
- This circuit is not optimized for low noise figure, it is a first step to a good design. The measured figures include losses of SMA-connectors and the relatively high loss of the microstrip lines on the epoxy-board. The use of teflon material would provide an improvement of ≈ 0.1 dB.

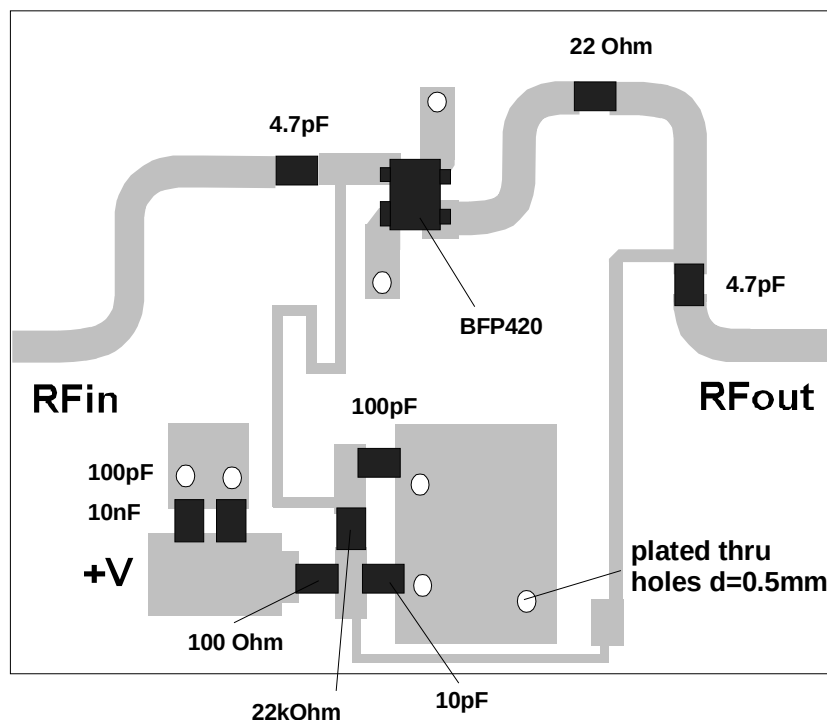
Resistor R2 is used to get higher rf-circuit-stability and return loss values at the output. It also affects the output intermodulation performance.

Component	Value	Unit	Size	Comment
R1	100	Ω	0603	bias / collector-resistance / $V_{R1} \approx 0.5V$
R2	22	Ω	0603	to improve stability and output return loss
R3	22	$k\Omega$	0603	bias / base-resistor
C1	4.7	pF	0603	inputmatch
C2	4.7	pF	0603	outputmatch
C3	100	pF	0603	rf-short
C4	10	pF	0603	outputmatch
C5	100	pF	0603	rf-short
C6	10	nF	0603	rf-short
Tr1			SOT343	SIEGET BFP420
TrL1				inputmatch and bias, $w=0.3mm$
TrL2				emitter-microstrip-line, $w=0.95mm$
TrL3				emitter-microstrip-line, $w=0.95mm$
TrL4				outputmatch, $w=0.95mm$
TrL5				outputmatch, $w=0.95mm$
TrL6				outputmatch and dc-bias, $w=0.3mm$
TrL7				inputmatch, $w=0.95mm$
Substrate	Fr4			$h=0.5mm$; $\epsilon_r=4.5$

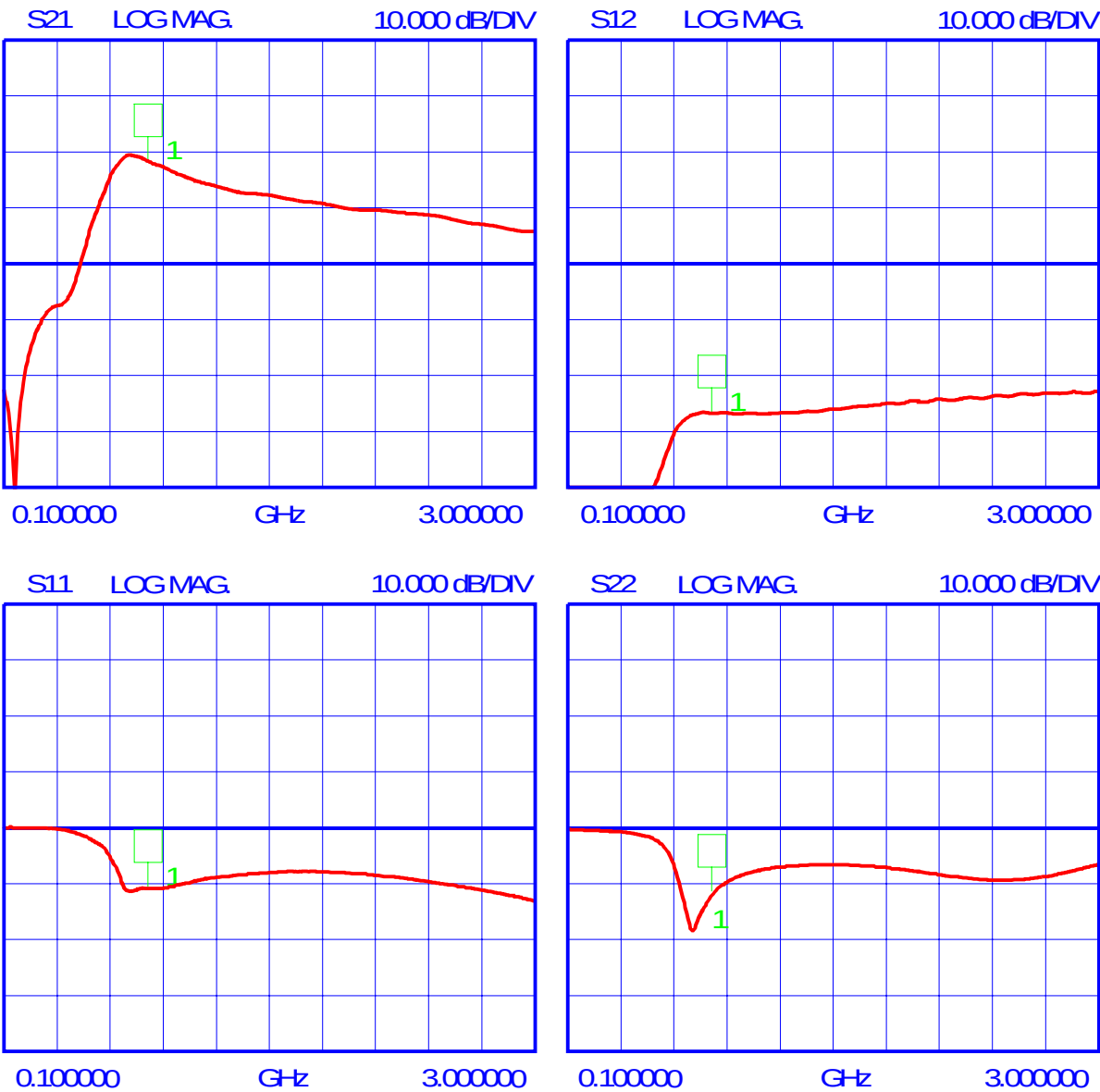
PCB Layout and Component Placement



scale 1:1
dim.: 25mm x 20mm



Measured data:



1 0.9 GHz

+

**Published by Siemens AG, Bereich Bauelemente,
Vertrieb, Produkt-Information, Balanstraße 73,
D-81541 München**

© Siemens AG 1997. All Rights Reserved

As far as patents or other rights of third parties are concerned, liability is only assumed for components per se, not for applications, processes and circuits implemented within components or assemblies.

The information describes the type of component and shall not be considered as assured characteristics.

Terms of delivery and rights to change design reserved.

For questions on technology, delivery and prices please contact the Offices of Semiconductor Group in Germany or the Siemens Companies and Representatives world-wide (see address list).

Due to technical requirements components may contain dangerous substances. For information on the type in question please contact your nearest Siemens Office, Semiconductor Group.

Siemens AG is an approved CECC manufacturer.